1. Create a worksheet that will calculate addition, subtraction, multiplication and division of 2 numbers provided by user.

2. Use Excel to solve the following mathematical problems:
   a) There are 25 female and 21 male students in each of 12 groups. How many students are there?
   b) There are 43 blue pens, 27 red pens and 56 black pens to be shared among 12 students. How many pens would each student get?
   c) 120 mini burgers were delivered for a buffet tea break. 15 were eaten by the helpers before the visitors (17 from China, 15 from India and 21 from Malaysia) arrive. How many mini burgers left for each visitor?

3. The duration of a long workshop program in US starts from 1st September 2023 until 30th August 2024. Use Excel to determine how many
   a) days,
   b) weeks,
   c) months and
   d) years
   within that time duration.

4. Given the following array of numbers.
   a) 3, 5, 7, 11
   b) 9, 5, 24, 12
   c) 10, 2, 14, 18, 7
   d) 27, 121, 25, 13, 58
   Find the minimum, maximum, average, greatest common divisor and the least common multiple of the numbers.

5. The employment rates of 6 countries from 2022 until 2024 are tabulated as below.

| No. | Employment Rates | Year | | |
|---|---|---|---|---|
| | | 2022 | 2023 | 2024 |
| 1. | Australia | 75.3 | 75.5 | 76.3 |
| 2. | Austria | 81 | 82 | 82.5 |
| 3. | Belgium | 74.4 | 76.4 | 77.9 |
| 4. | Canada | 78.3 | 79.2 | 79.9 |
| 5. | Czech Republic | 83.7 | 81.9 | 81.6 |
| 6. | Denmark | 83.4 | 84.4 | 84.3 |

   Use subtotal function in Excel to find sum and average of the employment rates with respect to year.

Assoc. Prof. Dr. Noor Fadiya Mohd Noor

6. Consider the following data.

| | EMPNO | ENAME | JOB | MGR | HIREDATE | SAL | COMM | DEPTNO |
|---|---|---|---|---|---|---|---|---|
| 1 | 7369 | Smith | Clerk | 7902 | 1980-12-17 00:00:00 | 800 | (null) | 20 |
| 2 | 7499 | Allen | Salesman | 7698 | 1981-02-20 00:00:00 | 1600 | 300 | 30 |
| 3 | 7521 | Ward | Salesman | 7698 | 1981-02-22 00:00:00 | 1250 | 500 | 30 |
| 4 | 7566 | Jones | Manager | 7839 | 1981-04-02 00:00:00 | 2975 | (null) | 20 |
| 5 | 7654 | Martin | Salesman | 7698 | 1981-09-28 00:00:00 | 1250 | 1400 | 30 |
| 6 | 7698 | Blake | Manager | 7839 | 1981-05-01 00:00:00 | 2850 | (null) | 30 |
| 7 | 7782 | Clark | Manager | 7839 | 1981-06-09 00:00:00 | 2450 | (null) | 10 |
| 8 | 7788 | Scott | Analyst | 7566 | 1987-04-19 00:00:00 | 3000 | (null) | 20 |
| 9 | 7831 | King | President | 23 | 1981-11-17 00:00:00 | 5000 | 23 | 10 |
| 10 | 7844 | Turner | Salesman | 7698 | 1981-09-08 00:00:00 | 1500 | 0 | 30 |
| 11 | 7876 | Adams | Clerk | 7788 | 1987-05-23 00:00:00 | 1100 | (null) | 20 |
| 12 | 7900 | James | Clerk | 7698 | 1081-12-03 00:00:00 | 950 | (null) | 30 |
| 13 | 7902 | Ford | Analyst | 7566 | 1981-12-03 00:00:00 | 3000 | (null) | 20 |
| 14 | 7934 | Miller | Clerk | 7782 | 1982-01-23 00:00:00 | 1300 | (null) | 10 |

Use Excel built-in function to find the
a) sum, average and the third largest values for column SAL.
b) sum, average and the second smallest values for column DEPTNO.

7. Given the following data.

| ID | Marital Status | Gender | Income | Children | Education | Occupation | Home Owner | Cars | Commute Distance | Region | Age |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 12496 | Married | Female | 40000 | 1 | Bachelors | Skilled Manual | Yes | 0 | 0-1 Miles | Europe | 42 |
| 24107 | Married | Male | 30000 | 3 | Partial College | Clerical | Yes | 1 | 0-1 Miles | Europe | 43 |
| 14177 | Married | Male | 80000 | 5 | Partial College | Professional | No | 2 | 2-5 Miles | Europe | 60 |
| 24381 | Single | Male | 70000 | 0 | Bachelors | Professional | Yes | 1 | 5-10 Miles | Pacific | 41 |
| 25597 | Single | Male | 30000 | 0 | Bachelors | Clerical | No | 0 | 0-1 Miles | Europe | 36 |
| 13507 | Married | Female | 10000 | 2 | Partial College | Manual | Yes | 0 | 1-2 Miles | Europe | 50 |
| 27974 | Single | Male | 160000 | 2 | High School | Management | Yes | 4 | 0-1 Miles | Pacific | 33 |
| 19364 | Married | Male | 40000 | 1 | Bachelors | Skilled Manual | Yes | 0 | 0-1 Miles | Europe | 43 |
| 22155 | Married | Male | 20000 | 2 | Partial High School | Clerical | Yes | 2 | 5-10 Miles | Pacific | 58 |
| 19280 | Married | Male | 20000 | 2 | Partial College | Manual | Yes | 1 | 0-1 Miles | Europe | 48 |
| 22173 | Married | Female | 30000 | 3 | High School | Skilled Manual | No | 2 | 1-2 Miles | Pacific | 54 |
| 12697 | Single | Female | 90000 | 0 | Bachelors | Professional | No | 4 | 10+ Miles | Pacific | 36 |
| 11434 | Married | Male | 170000 | 5 | Partial College | Professional | Yes | 4 | 0-1 Miles | Europe | 55 |
| 25323 | Married | Male | 40000 | 2 | Partial College | Clerical | Yes | 1 | 1-2 Miles | Europe | 35 |
| 23542 | Single | Male | 60000 | 1 | Partial College | Skilled Manual | No | 1 | 0-1 Miles | Pacific | 45 |
| 20870 | Single | Female | 10000 | 2 | High School | Manual | Yes | 1 | 0-1 Miles | Europe | 38 |
| 23316 | Single | Male | 30000 | 3 | Partial College | Clerical | No | 2 | 1-2 Miles | Pacific | 59 |

Assoc. Prof. Dr. Noor Fadiya

Find the total income for
a) married individuals.
b) female individuals.
c) Europe region.
d) Single female.
e) Pacific male.
f) Europe male with professional occupation.
g) Married male with partial college education background.
h) Europe partial college background between the age of 40 and 61.

8. Use Excel to find sum of the series:
   a) $1 - x^2 + x^4 - x^6 + .. + x^{40}, x = 1.447$
   b) $2 + 4x + 8x^2 + .. + 65536\, x^{15}, x = 0.78$
   c) $\sum_{i=0}^{65} 3i\ (2.53)^{i\left(\frac{1}{2}\right)^i}$

9. Find all the trigonometric and hyperbolic functions of the angles $\frac{\pi}{5}$, $30^0$ and $215^0$.

10. Generate 10 random values between -100 and 100 in a column. Rewrite the values in another column by adding the decimals 0.412, 0.333, 0.54, 0.91, 0.128, 0.6667, 0.0003, 0.0247, 0.102, 0.439 to the values respectively. Next round up and round down the non-integer values to the nearest integers.

Assoc. Prof. Dr. Noor Fadiya Mohd Noor